

Fault-Tolerant Distributed Training of Language Models via Avian Message Passing Interfaces

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I. INTRODUCTION

Modern artificial intelligence applications have been enabled by the development of large artificial neural networks that benefit from empirical scaling laws which incentivize models of ever increasing size. However, as the size of models have grown, it has become necessary to leverage growing numbers of parallel processors. The extent of model growth has made it such that training can no longer take place within a single facility. As we approach artificial general intelligence, it will become the case that the models are so large that computation exceeds that of what is available in a single data center – requiring innovations in algorithmic methods and system design [?]. At this scale, novel challenges emerge in order to (1) coordinate computation, (2) minimize inter-facility communication overhead, and (3) ensure that both the computation and communication systems are fault-tolerant.

Unfortunately, due to the complexity of the underlying parallel computing hardware and hierarchical network topology, developing infrastructure for distributed systems and training infrastructure at-scale remains an open challenge for the research community. We hypothesize that the scale and inefficiency of modern deep learning systems has grown so large that we can instead leverage biological neural networks as drop-in replacements for key components of state-of-the-art machine learning distributed training systems.

Previous works have sought to estimate costs via models of kernel execution time; however these are idealized settings. In this work, we develop a model based on an ensemble of biological neural networks based on existing agent architectures.

As is common in modern distributed training of neural networks, we consider a computing topology in which large model training is coordinated across multiple nodes of GPU accelerators. In particular, we examine the setting in which model size is sufficiently large such that accelerators are distributed across multiple computing facilities (i.e. data centers); and communication of data is required over large real-world physical distances, a setting known to exhibit



Fig. 1: A common misconception is that birds draw power from the electric grid via wire perching. In fact, birds run on power derived from a *biological* process of digestion.

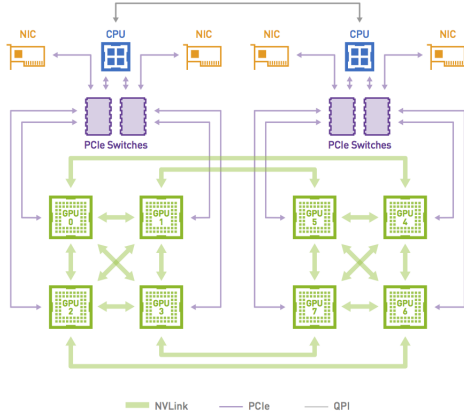
packet loss. In this regime, it is not possible to utilize the high-bandwidth, high-speed networking infrastructure high-performance computing clusters (e.g. NVLink, Infiniband, ROCE); instead data must be transferred utilizing standard communication protocols over the internet. However, these methods for data transmission are known to be financially costly, power-intensive, and lossy. Can we do better?

To address these concerns, we develop an organic, low-power approach for data transmission in large-scale distributed systems utilizing avian-based communication protocols, which we refer to as Avian Message Parsing (AMP; See Figure 2). Birds are notorious for requiring minimal electricity relative to their electrical and optical telecommunication-based counterparts (Fig 1); and are capable of supporting the weight of physical memory hardware (e.g. a raven is capable of carrying a hard disks; as described in Section ??).

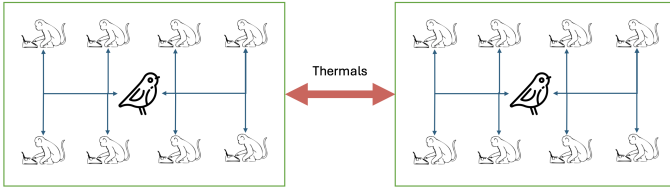
II. PROPOSED METHOD: AVIAN MESSAGE PARSING

We introduce Avian Message Parsing (AMP), a biologically inspired replacement for electrical and optical networking infrastructure. In a standard distributed system, we replace electrical and optical inter-facility interconnects with avian messengers (i.e. birds) which travel between computing units and facilities via flight; bypassing bottlenecks and sub-optimal routing required by land-based infrastructure. In our proposed architecture, data center operators and ornithologists attach hard disks with model predictions and intermediate activations

* Equal Contribution. Author order determined by highest score in Flappy Bird.



(a) Traditional GPU topology for distributed computation which leverages costly electrical infrastructure for data transmission.



(b) Proposed Avian Message Parsing which utilizes *organic, low-power* data transmission for increased efficiency deep learning training.

Fig. 2: Comparison of GPU-GPU communication with standard electrical and improved bird-based infrastructure.

to trained birds at data centers; which then fly to sister data centers carrying other components of the model parameters and computation; where receiving operators integrate the received information into their compute infrastructure. Furthermore, we direct our avian messengers to use existing convection patterns in the Earth’s atmosphere to accelerate transmission speeds. Under this configuration, data centers no longer need to be densely co-located in a single region (e.g. Northern Virginia), a practice known to induce strain on local electrical grid infrastructure; potentially increasing the electricity costs to rate-payers. Instead, data centers can be placed freely along wide swaths of regions along naturally occurring thermal cells and take advantage of renewable resources and wind patterns.

A. Transfer to persistent storage

Unfortunately, there are no power outlets in the sky so it necessary to offload data to persistent storage such as solid-state and hard disks as it is impossible for GPUs to retain the values of weights and activations without power. While it would be ideal to use cloud storage to write and retrieve data, such technology does not exist as clouds are made of water and are incapable of holding the weight of physical hard disks.

As a result, it is necessary for technicians to offload data from VRAM to persistent memory storage devices to birds at computing facilities.



Fig. 3: The ring-billed gull, widely considered to be the first to implement ring attention in the wild. Its effective data transfer speed is approximately faster than a snail.

B. Bandwidth and data weight

Birds are limited by their small little wings and thus cannot carry all of the weights required.

Luckily, people have spent a lot of time thinking about what could reasonably be strapped onto birds for a *different* type of science. Biologists regularly affix transponders onto wild birds to track nesting locations, migration patterns, survival rates, and other community statistics.

A common choice is to limit transponder weight to $\leq 5\%$ of the bird’s body weight (or the mean body weight of that species). [?]

However, transponder weights are designed to be minimal so that birds are not affected by the device’s placement and with the understanding that birds may carry other loads (such as nesting materials or prey) while the transponder is carried. Since the data transfers are relatively short (often only a few hours to days) and transient, we can allocate much heavier loads.

To find a reasonable load value, we consider homing pigeons. Homing pigeons are frequently cited as carrying up to 75g with training, when they only weigh 315-425g This is a load-to-weight ratio of 0.1875, for a 400g bird.

C. Directionality of data transfer

In contrast to traditional networking communications that assume bandwidth speeds are constant regardless of directionality, AMP encounters real-world physical limitations in regards to the transmission topology of data.

1) *Ring Attention*: Unfortunately, avian message transmitters are known to exhibit a condition known as “bird-brain” in which their spatial senses are limited; and it is a commonly observed phenomena in which birds are only able to home in a single direction effectively.

2) *Thermals*: In our proposed architecture, we leverage naturally occurring thermals, air pathways of high velocity, which enable faster flight by our transmitters. However, these

3) *Existing migratory pathways*: We can limit the amount of training necessary by working along existing migration paths.

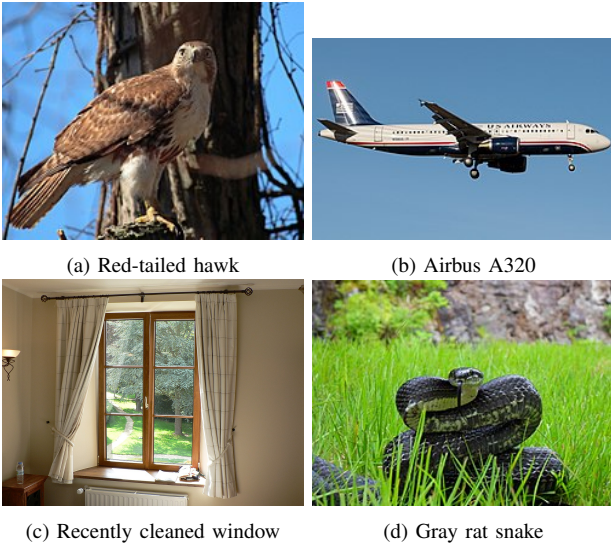


Fig. 4: Packet loss in AMP can be caused by a number of environmental factors.

D. Data transfer speed

Bandwidth is measured in FLAPs (flown logs by aviation post, per second). To measure bandwidth, we must look at the flying speed of birds and their distance per day covered.

E. Packet loss

All data transfers incur some risk of data loss. AMP is robust to power grid failure, civil unrest, road closures, and most minor solar flares;¹ However, AMP also introduces new potential sources of packet loss. Long-haul bird flight carries a number of inherent risks. Figure 4 demonstrates four common environmental hazards for small birds.

As a proxy for expected packet loss rates on AMP, we consider a more well-studied bird transit pattern: seasonal migration. Seasonal migration is not a perfect proxy— migrations are much longer trips than datacenter transfers, typically performed in weather that is marginal for that bird’s flight, and without safe overnight nesting locations. Despite this, many species have quite low migration loss rates [?].

We take as an example the Greater Snow Goose, which has a particularly punishing autumn migration from the North Arctic, often featuring freezing temperatures. While the exact staging locations during this part of the snow goose migration are not well understood, this is a migration of no less than 3,000km from Bylot Island to a St. Lawrence River estuary, undertaken in approximately 5 weeks [?]. goosepath state a monthly survival rate of adult Greater Snow Geese during the migration period of 98.9%.² Given these numbers, we compute an approximate per-km expected loss of 4.58×10^{-6} birds. Note that because of the particularly difficult conditions of

this migration and the likelihood that the geese take an indirect path much longer than 3,000 km, this should be considered a very weak lower bound on packet loss.

How does this compare to existing systems? In some situations (e.g. voiceover), packet loss up to 5-10% may be acceptable [?]. An acceptable Ethernet packet loss rate has elsewhere been listed as 1-2% Using the lower bound calculated above, we can estimate that goose AMP over distances less than 21,818km will have a less than 1% packet loss rate; this means that goose data transfer over *half the circumference of the Earth* has an acceptably low packet loss rate.

1) *Bird vs. flock failure rates:* Of course, we care not just about total packet loss rate but the grouping of packets lost. Because we are considering each bird to be a “packet,” the size of birds chosen impacts the amount of data lost if a single bird is lost.

2) *Reduction of packet loss via cross-bird data redundancy:* The above analysis assumes that each bird carries completely unique data. Of course, in practice, some redundancy in the system is advised. We recommend sharing the desired data to transfer across all birds in the flock by maintaining two copies of each data point, distributed to two different birds. Rather than making duplicates of each physical storage device’s contents, we share *sub-packets* of smaller data units, e.g. weights for an individual matrix. Each bird’s data storage device must also maintain a lightweight *registry* of IDs for weight matrices in the total data transfer.

This has two functions: first, it reduces the likelihood of any catastrophic data loss event. Second, it also increases FLAPs over the total data transfer: because the data is shared across all birds and we maintain a listing of the total data expected within each bird’s storage, it is trivial to verify when all data has arrived, even if not all birds have yet arrived. This means that, on average, the data transfer can be considered complete when *half the flock has arrived*, reducing the odds of slowdown because of a lost, delayed, or simply lazy bird.

III. LIMITATIONS

We believe our proposed system provides a strong model for estimating the cost and efficiency of machine learning systems. Unfortunately, its feasibility suffers because of several technological limitations of our time.

Most data storage devices, strangely, are not well-suited for transport in cold, potentially wet conditions, despite being regularly used for “cloud storage.” This discrepancy is not well-understood; at least, we emailed Sandisk and they didn’t have any answers for us.

Additionally, our work relies on a strong assumption that birds are real, which has been recently called into question in the scientific community [1].

REFERENCES

- [1] Shoemaker, Lauren. “Birds Aren’t Real.” *Avian Aesthetics in Literature and Culture: Birds and Humans in the Popular Imagination* (2022): 215.

¹ A sufficiently strong solar flare, however, can cause packet redirection [?].

² The juvenile monthly survival rate, on the other hand, is an astonishingly low 66.2%. For the purposes of this work, we assume that AMP is not using child bird labor.